

## **Pro-drop in Brazilian Portuguese**

Universität Konstanz  
Grammar Development  
Summer Semester 2026  
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Speech and Language Processing  
August 20th, 2026

## 1 Introduction

The present work is a result of the development of a small grammar on XLE on pro-drop in Brazilian Portuguese. It consists of grammatical rules and a lexicon pertaining to the possible (and impossible) omission of subject pronouns in specific linguistic contexts in the language. The grammar was built from a collection of sentences, grammatical and ungrammatical, that are parsed accordingly.

Brazilian Portuguese is a partial pro-drop language (i. e. it offers the possibility of null subjects and objects), and that has much to do with how it has evolved away from European Portuguese, which is still considered a full pro-drop language alongside other Romance languages like Spanish and standard Italian. The decrease in use of null subjects in BP is a direct consequence of the loss of verbal morphology the language has undergone overtime, as shown by Duarte (1993, 1995, 2000), from six forms to ultimately only three in everyday uses (in both written and spoken language). The changes equate to the decreased use of the second person singular (*tu*, replaced by *você*, formerly a third person term of address), first person plural (*nós*, replaced by *a gente* [*the foks*], formally a third person singular expression that now also includes the speaker) and second person plural (*vós*, completely eradicated in BP and replaced with *vocês*, formerly a third person term of address). For that reason, referential null-subjects in out-of-the-blue subjects are not allowed (e.g. *\*Comprou pão* [bought bread]) .

Due to these losses BP cannot have the same possibilities for null subjects as its romance counterparts. However one that remains is the possibility for null subjects in embedded clauses if some locality constraints are obeyed, in which case it is anaphoric to a c-commanding NP (e.g. *Ninguém acha que [e] é estúpido* [nobody thinks that [e] is stupid]). Failing to obey the locality constraints creates an ungrammatical sentence (*Mário disse que Ana acha que [e] é feio* [Mário said that Ana thinks that [e] is ugly.]).

## 2 Grammar implementation

The sentences that served as a basis for the implementation of the grammar rules and lexicon are the following:

Positive examples:

- 1) Ele comprou pão. (*He bought bread.*)
- 2) Paulo disse que ele comprou a casa. (*Paulo said that he bought the house.*)
- 3) Paulo disse que comprou a casa (*Paulo said that [e] bought the house.*)

- 4) Ninguém disse que ele é estúpido. (*Nobody said that he is stupid.*)
- 5) Ninguém disse que é estúpido. (*Nobody said that [e] is stupid.*)
- 6) João comprou pão e o levou para casa. (*João bought bread and took it home.*)
- 7) João comprou pão e levou para casa. (*João bought bread and took [e] home.*)
- 8) Mário disse que Ana acha que ele é feio. (*Mário said that Ana thinks that he is ugly.*)
- 9) Elas são simpáticas. (*They are nice.*)
- 10) Vamos para casa? (*Let's go home?*)

#### Negative examples

- 1) \*Comprou pão. (*[e] Bought bread.*)
- 2) \*Mário disse que Ana acha que é feio. (*Mário said that Ana thinks that [e] is ugly.*)
- 3) \*São simpáticas. (*[e] are nice.*)
- 4) \*Nós vamos para casa? (*Let's go home?*)
- 5) \*Vai para casa? (*Can you go home?*)

The first rules implemented included a rule for sentence (S), with an embedded mandatory NP as the subject and VP. The VP rule included a mandatory verb and optional NP as an object, while the NP rule included an optional determiner and a mandatory noun, both of which could be substituted by a pronoun.

$S \rightarrow NP: (^ SUBJ) = !$

$(! CASE) = \text{nom};$

$VP: ^=!$ .

$VP \rightarrow V: ^=!$ ;

$(NP: (^ OBJ) = !$

$(! CASE) = \text{acc}).$

$NP \rightarrow \{(D)$

$N$

$| \text{PRON}\}.$

Then, to allow for embedded clauses, the CP rule was added. To accommodate for the possibility of an anaphoric null subject, we also added a second sentence rule (S2) and placed it after the complementizer in the CP rule.

$VP \rightarrow V: \wedge=!$ ;  
 $(NP: (\wedge OBJ) = !$   
 $(! CASE) = acc)$   
 $(CP: (\wedge COMP) = !).$

$CP \rightarrow C$   
 $S2.$

$S2 \rightarrow \{e: (\wedge SUBJ PRED) = 'pro'$   
 $| NP: (\wedge SUBJ) = !$   
 $(! CASE) = nom\}$   
 $VP.$

The possibility of a predicate was also introduced in the VP rule for examples including a connecting verb and an adjective phrase, as well as a rule for the adjective phrase itself.

$VP \rightarrow V: \wedge=!$ ;  
 $(\{NP: (\wedge OBJ) = !$   
 $(! CASE) = acc$   
 $| AP: (\wedge PREDICATE) = !$   
 $(! CASE) = nom\})$   
 $(CP: (\wedge COMP) = !).$

$AP \rightarrow A.$

In Portuguese, an object that takes the form of a pronoun often antecedes the verb, therefore the optional rule was added to the VP:

$VP \rightarrow (NP: (\wedge OBJ) = !$   
 $(\wedge PRON-TYPE) = pers$   
 $(\wedge NTYPE) = pron$   
 $(! CASE) = acc)$   
 $V: \wedge=!$ ;  
 $(...)$

For the more complex embedded clauses (with one CP inside another), the solution was to create two different rules for CP: one to account for the more flexible disposition described in earlier sentence examples, where the NP could be present or omitted, and another one for a segment in which the c-command relations require an overt subject.

CP2 → C  
S.

Finally, we implemented a rule to account for interrogatives, specifically suggestions which, in BP, force a null subject. To accomplish that, we restricted the rule to apply only to third person plural, in the present tense and in the presence of an interrogation mark.

Sint --> e: (^ SUBJ PRED) = 'pro'  
(^ SUBJ PERS) = 1  
(^ NUM) = pl;  
VP: (^ TNS-ASP) = pres;  
INTER.

## References

- Holmberg, A.. (2010). Parameter Variation: Null Subjects in Minimalist Theory. Biberauer, T., Holmberg, A., Roberts, I., Sheehan, M. (Eds.), *Null Subject Parameters* (pp. 88-124). Cambridge.
- Kato, M. A.. Strong and weak pronominals in the null subject parameter. *Probus*, 11(1):1-38. 10.1515/prbs.1999.11.1.1
- Saab, A.. (2016). The Morphosyntax of Portuguese and Spanish in Latin America. Kato, M. A.; Ordóñez, F. (Eds.), *On the notion of partial (non-)pro-drop in Romance* (pp. 49-7). Oxford Studies in Comparative Syntax.

## APPENDIX A

### Gloss of the examples from the testsuite

- (1) *Ele comprou pão.*  
he bought bread  
'He bought bread.'

- (2) *Paulo disse que ele comprou a casa.*

Paulo said that he bought.3SG the house  
'Paul said that he bought the house.'

(3) *Paulo disse que [e] comprou a casa.*  
Paulo said that [e] bought.3SG the house  
'Paul said that [e] bought the house.'

(4) *Ninguém disse que ele é estúpido.*  
nobody said.3SG that he is.3P.SG stupid.MASC.SG  
'Nobody said that he is stupid.'

(5) *Ninguém disse que [e] é estúpido.*  
nobody said.3SG that [e] is.3SG stupid.MASC.SG  
'Nobody said that [e] is stupid.'

(6) *João comprou pão e o levou para casa.*  
João bought.3SG bread and it.ACC took.3SG to home  
'João bought bread and took it home.'

(7) *João comprou pão e [e] levou para casa.*  
João bought.3SG bread and [e] took.3SG to home  
'João bought bread and took [e] home.'

(8) *Mário disse que Ana acha que ele é feio.*  
Mário said.3SG that Ana thinks.3SG that he is.3SG ugly.MASC.SG  
'Mário said that Ana thinks that he is ugly.'

(9) *Elas são simpáticas.*  
they.FEM.SG are.3PL nice.FEM.PL  
'They are nice.'

(10) *Vamos para casa?*  
*go.1PL. to home*  
'Let's go to the store?'

- (11) \**Comprou pão.*  
bought.3SG bread  
Intended: ' [e] bought bread.'
- (12) \**Mário disse que Ana acha que [e] é feio.*  
Mário said.3SG that Ana thinks.3SG that [e] is.3SG ugly.MASC.SG  
Intended: 'Mário said that Ana thinks that [e] is ugly.'
- (13) \**[e] São simpáticas.*  
[e] are.3PL nice.FEM.PL  
Intended: '[e] are nice.'
- (14) \**Nós vamos para casa?*  
we go.1PL to home  
Intended: 'Let's go home?'
- (15) \**Vai para casa?*  
go.1PL to home  
Intended: 'Can you go home?'